Yuze Fu

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Website | LinkedIn | Github

Research Interests

Numerical Methods for Partial Differential Equations, Physics-based Computer Graphics

Education

- **University of Toronto Scarborough** — *Sep 2023 - June 2027 (expected)*
  Honours Bachelor of Science - Mathematics and Computer Science — Toronto, Ontario, Canada
  - GPA: 3.99/4.00
  - Math Courses: Abstract Algebra, Numerical Methods for PDEs, Number Theory, Graph Theory, Analysis
  - CS Courses: Machine Learning, Operating System, Database, Human-computer Interaction
- **High School Affiliated to Fudan University** — *Sep 2020 - June 2023*
  High School Diploma — Shanghai, China
  - Studied Math, Physcis, Chemistry, Geography
  - Member of the school's competitive programming team for the Informatics Olympiad
  - Member of the school's soccer team, playing against other high schools in Shanghai

Research

- **Adaptive Particle Splitting for Fluid Simulation** — *May 2026 - Present*
  Supervised by Eitan Grinspun and Jonathan Panuelos — Dynamic Graphics Project Lab
- **Automated Misinformation Detection & Claim Verification Pipeline** — *Jun 2025 - Aug 2025*
  Research Assistant — Rotman School of Management
  - Developed a two-phase NLP pipeline using SBERT and LLMs to classify social media misinformation.
  - Applied cosine similarity and embedding-based filtering to align thousands of tweet-article pairs.
  - Leveraged DeepSeek API to classify claim relationships as identical, contradictory, or unrelated.
  - Optimized system accuracy via iterative threshold tuning and incremental data processing strategies.

Teaching

- **Teaching Assistant for CSCB09 (System Programming)** — *Winter 2026*
- **Teaching Assistant for CSCB36 (Introduction to the Theory of Computation)** — *Fall 2025*
- **Teaching Assistant for MATA22 (Linear Algebra I for Mathematical Science)** — *Winter 2025*
- **Teaching Assistant for CSCA67 (Discrete Mathematics)** — *Fall 2024*

Honors and Awards

- **Department of Computer Science Award** — *2026*
  University of Toronto — $12000
  - 16-Weeks Undergraduate Student Research Award funded by the Department of Computer Science.
- **University of Toronto Scholar** — *2024*
  University of Toronto — $1500
  - University-wide in-course scholarships recognized for most outstanding academic achievements.
- **Dean's List** — *2024, 2025, 2026*
  University of Toronto Scarborough
- **Second Prize in the National Olympiad in Informatics (NOIP2020)** — *2020*
  China Computer Federation

PROJECTS

- **Classification of Finitely Generated Modules over PIDs** *Mar 2026 - Apr 2026*
  Abstract Algebra Essay
  - Classified finitely generated modules over PIDs to unify the structure of abelian groups and operators.
  - Formulated the bijection between $F[x]$-modules and linear maps to prove operator structural identities.
  - Constructed Rational and Jordan Canonical Forms using Invariant Factor and Elementary Divisor forms.
  - Concluded with a review of the theoretical and computational trade-offs between different matrix canonical forms.
- **Numerical Analysis of Finite Difference Schemes for Hyperbolic PDEs** *Feb 2026 - Apr 2026*
  Numerical Methods for PDEs Projects
  - Implemented 8 finite difference schemes using vectorized NumPy operations, optimizing performance for advection-based simulations.
  - Evaluated the stability and order of convergence for standard schemes (e.g. Lax-Wendroff, Leapfrog) under varying CFL conditions.
  - Developed SBP-SAT boundary treatments for high-order Runge-Kutta methods, ensuring stable and consistent convergence up to the 4th order.
  - Analyzed theoretical vs. experimental trade-offs between ghost cell extrapolation and SBP-SAT operators for boundary stability.
- **NSI Forecasting: Machine-Learning-Driven Neighborhood Safety Index** *Nov 2025 - Dec 2025*
  Machine Learning Project
  - Engineered a custom Neighborhood Safety Index (NSI) based on the Canadian Crime Severity Index to quantify urban safety levels.
  - Evaluated Ridge, KNN, and FNN models to compare baseline linear accuracy against non-linear neural approximations.
  - Developed an LSTM network achieving a leading R^2 of 0.8295 by modeling complex 12-month temporal sequences.
  - Optimized performance through grid search tuning of architecture depth, activation functions, and k-neighbors.

COMMUNITY INVOLVEMENT

- **UTSC Accessability Services** *Sep 2024 - Present*
  Volunteer Notetaker
  - Produced comprehensive lecture and tutorial notes to support students with diverse accessibility needs.
  - Courses: Probability (STAB52), Computation Theory (CSCB36), Group Theory (MATC01), Numerical Methods for PDEs (MATD10)
- **UTSC Summer School** *Jul 2025 - Aug 2025*
  Teaching Assistant
  - Helped Professors to hold seminars about computer science topics for high school students.
  - Answered students' questions and provided information about university lives.
- **The Association of Mathematical and Computer Science Student (AMACSS)** *Sep 2024 - Apr 2025*
  Course Representative
  - Developed review materials and facilitated pre-exam seminars to improve peer academic performance.
  - Courses: Discrete Math (CSCA67), Introduction to Computer Science (CSCA08)
- **UTSC Google Developer Group (GDG)** *Oct 2024 - Apr 2025*
  Academic Director
  - Supported the execution of "Build with AI," an industry-focused event with 500+ registrations and live AI demos.
  - Created an interactive emotion-matching game that successfully increased participant engagement.

SKILLS AND HOBBIES

- **Programming Languages:** C/C++, Python, Assembly, SQL, Shell Script, Java, R, LaTeX, Markdown
- **Frameworks and Librarys:** pandas, NumPy, SciPy, PyTorch, scikit-learn
- **Developer Tools:** Houdini, Docker, PostgreSQL, Git
- **Hobbies:** Soccer, Piano, Swimming, Traveling, Movie, RPG Game